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Review Classmates: Part 5: Modeling Credit Card Default Risk and Customer Profitability

Review by December 16, 11:59 PM PT

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Comparing the Binary classification and regression credit prediction approaches

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December 10, 2015

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What was the binary classification model you developed in Final Quiz Part 1, Question #1? Type your model in the box below.

= [credit debt + autodebt+ age+years-at-employer+income]/[5]

Please refer to the Discussion for Peer Assessment document (<https://d396qusza40orc.cloudfront.net/phoenixassets/Jackie%27s%20folder/Discussion%20for%20Peer%20Assessment.pdf>) and the Peer Review Comments and Discussion (https://d396qusza40orc.cloudfront.net/phoenixassets/Jackie%27s%20folder/Final_Peer-Review-Discussion_10-30-2015.pdf) document during the review process.

Does the submitted model include at least 2 of the 6 pieces of data about the credit card applicants (age, years at current employer, years at current address, income over the past year, current credit card debt, and current automobile debt)?

☒ 2 pts

Yes

☐ 1 pt

No, the model only includes 1 of the pieces of data

☐ 0 pts

No model was provided, or the model includes none of the pieces of data

At the threshold score for the training set, what is the cost per event on the test set? Refer back to Final Project Quiz 1, question 6 for your answer.

Note: if you find that your costs per event on the test set are much higher than your costs per event on the training set, you should consider making your model simpler – probably using fewer input variables – as it is probably still over-fitting the training set data. Problems that are were not obvious at the ROC-curve stage may emerge when minimizing costs.

1250

Is the minimum cost per event submitted by your peer an acceptable cost per event?

☐ 0 pts

Cost per event > \$1,250 (no incremental value)

☒ 1 pt

Cost per event > \$1,000 to <=\$1,250

☐ 2 pts

Cost per event >\$800 to <= \$1,000

☐ 3 pts

Cost per event \$600 to <=\$800

☐ 0 pts

Cost per event <\$600 (unrealistic to expect that any model could produce a cost per event this low)

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Your boss asks you to recommend a standard procedure for selecting future credit card applicants. Based on internal data alone (*not using the Eggertopia scores*) and the test set performance of your two models (the binary classification model from Final Project Quiz 1 and the linear regression model from Final Project Quiz 4), which method would you recommend - binary classification, linear regression or some combination or modification of the two? Explain why

you recommended what you did, referencing at least two topics/points from the course.

The binary classification as a default prediction model as it is based on an index that has been drawn from other factors, hence seems to me simply to apply. The problem, that is according to me is that, the binary classification model still leaves a lot of uncertainty. One big point the binary classification model has is that it enables us to determine the cost per event. However, there is no underlying scientific model behind the process, making it difficult to reproduce and replicate. This is because lacking a model to use as reference, the binary classification model is a trial and error process, If I can put it that way. As long as the permutations of factors used to measure creditworthiness can be as many as one wants, uncertainty still lurks in the performance of binary classification results we obtain. Moreover it must be noted that based on the confusion matrix the difficulty of enhancing model performance through higher sensitivity (positive predictive value/true positive rate), while at increasing model specificity (Negative predictive value/true negative rate) is not easy, leading to the need for a trade off between the two binary classification model performance measures. In other words, increasing the extent to the classification model does not contain false positives per test incidence), is achieved with trade off of increasing false negatives). This implies that a threshold has to be determined that may not be the best but just optimal. In this case, we determined that the training set cost per event was \$963 per event, which for obvious reasons rose to \$975 per event in the test set (both at 0.70 AUC). The problem is that a yes or no classification system that therefore simplifies the complexity of the credit decision making system. This was clearly noted in the lecture when we were informed that based on the binary classification system, it happens that it is not easy at the outset to determine whether or not, clients that do not pause any default problem on providers of credit end up generating profits for them. On the contrary, clients that are potentially bad for the credit provider (have high likelihood of defaulting on loans they receive, ironically end up generating higher profitability through higher interest payments. In any case, pricing the contribution of binary classification model becomes difficult as it often assumes as if prior to its presence that was no information available such that the contribution it makes is often inflated. It is therefore true that a client can contribute to lowering the cost of credit disbursement but may not necessarily contribute significantly to profitability. This is where the regression prediction model comes in. The regression model, involves the direct use of the factors (variables) that impact on performance on an debtor of a loan contracted as viewed from the vantage point of profitability, which embodies the cost of the transaction as well). Regression model approaches the problem from the vantage point of profitability. Based on the performance of the linest model, we can with higher degree of certainty gauge the performance of an individual debtor. There is more precision in predicting outcomes as we have the leverage to improve on the model performance to parsimonious level. In this case for example, we can use income, age, and years at employer as predictors of prospective client. The problem with the regression model however is that the factors for which the data we have at hand, can only explain less than 80 percent of the variance in profitability, leaving more than 30 percent uncertainty. Another problem is that we can not use the linest model that generates the betas for the training set to determine whether or not the performance is better or worse on the test set (simply because the linest model just estimates again the betas, which for obvious reasons will not be any different in the test model compared with those generated in the training model. This is why to gauge model performance we resorted to calculating mean mean residual squares from difference between actual profitability values and estimates profitability values)squared) divided by degree of freedom. This enabled us to estimate the mean root square error that is used as a measure of the extent to which the estimated values are closer to actual values. Given the reality that both prediction models have their strengths and weaknesses, I would argue for a combination of both for best results. Binary classification should help in fleshing out factors that are by nature not easy to quantify, while the regression model should handle the brute force, scientific modelling credit worthiness estimation process. The process should be complementary rather than substitute of one another for best results.

Does the learner provide a clear recommendation for which model (or combinations of models) to use to select future credit card applicants?

- ☒ 2 pts
Yes, there is a clear recommendation
- ☐ 1 pt
There seems to be a vague recommendation, but it is not clearly or definitively stated
- ☐ 0 pts
No, there is no recommendation at all

No matter which model(s) is recommended, does the learner provide an explanation for their choice which references at least two topics/points from the course?

- ☒ 3 pts
Yes, the learner mentions at least two topics/points from the course, and includes the idea that the best model is the one with the lowest cost per event
- ☐ 2 pts
Yes, the learner explains and mentions at least two topics/points from the course, although does not explicitly mention that the best model is the one with the lowest cost per event
- ☐ 1 pt
No, the learner only references one topic/point from the course
- ☐ 0 pts
No, the learner provides no explanation or her/his explanation has no connection to the course materials

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